Ely Giancoli Ferreira de Miranda

Postdoctoral Researcher · Atomic and Molecular Physics

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Personal Profile

I hold a PhD in Physics from the Institute of Physics of the University of São Paulo and work in Atomic and Molecular Physics, with emphasis on Computational Physics and Theoretical Chemistry. My work is on the development and application of theoretical methods for electron-molecule scattering, with focus on resonant anionic states and dissociative processes in organic systems. Over the past years I have been connecting two traditions that are usually kept apart: quantum scattering theory, which tells us how a transient anion is formed, and nonadiabatic dynamics with correlated methods, which tells us what the molecule does next. I am currently a postdoctoral researcher at Unicamp, in the group of Prof. Arnaldo Naves de Brito.

Education and Positions

Postdoctoral Researcher 2026 – present
University of Campinas (Unicamp), Campinas, Brazil FAPESP Grant #2026/07066-7

Supervisor: Prof. Arnaldo Naves de Brito, Institute of Physics "Gleb Wataghin"

PhD in Physics (Molecular Physics) 2021 – 2026
University of São Paulo (USP), São Paulo, Brazil FAPESP Grant #2021/09837-7

Thesis: Models for Transient Anion Dynamics
Advisor: Prof. Márcio T. do N. Varella
Low-energy electron scattering by radiosensitizers, quantum scattering theory (Schwinger Multichannel Method), dissociative electron attachment, nonadiabatic dynamics with multireference methods

Research Exchange 2024 (6 months)
Aix-Marseille Université, Marseille, France FAPESP BEPE Grant #2023/14792-8

Supervisor: Prof. Mario Barbatti
Nonadiabatic molecular dynamics with fully correlated methods (CASPT2, MRCI, L-PDFT, MRSF-TDDFT). Interfaces between Newton-X and OpenMolcas for photochemical processes and dissociative electron attachment.

M.Sc. in Physics 2019 – 2020
University of São Paulo (USP), São Paulo, Brazil CNPq Fellowship

Dissertation: Anionic states of resveratrol and its subunits • **Advisor:** Prof. Márcio T. do N. Varella
Quantum scattering calculations (Schwinger Multichannel Method), elastic cross sections, resonance characterization

B.Sc. in Physics 2015 – 2018
University of São Paulo (USP), São Paulo, Brazil

Publications

Six articles in peer-reviewed international journals. Full record on ORCID.

- **E. G. F. de Miranda** and M. T. do N. Varella, "Anion states and dissociation pathways of the universal radiosensitizer model 2,4-dichloro-5-nitropyrimidine"
  The Journal of Physical Chemistry A · 10.1021/acs.jpca.6c00754 2026
- M. Barbatti *et al.*, including **E. G. F. de Miranda** (30 authors), "The Newton-X platform for mixed quantum–classical dynamics"
  Physical Chemistry Chemical Physics · 10.1039/D6CP01391K 2026

- P. Guerra, M. Mendes, **E. G. F. de Miranda**, R. Rodrigues, L. M. Cornetta and F. Ferreira da Silva, "The role of benzonitrile chlorination explored by dissociative electron attachment"
  ChemPhysChem · 10.1002/cphc.70406 2026
- **E. G. F. de Miranda**, R. S. Mattos, S. Mukherjee, J. M. Toldo, C. H. Choi, M. T. do N. Varella and M. Barbatti, "Surface Hopping with fully correlated methods"
  Journal of Chemical Theory and Computation · 10.1021/acs.jctc.5c01529 2026
- H. Abdoul-Carime, **E. G. F. de Miranda** and M. T. do N. Varella, "Low-energy (0–9 eV) electron interaction with gas phase 1,3-dichlorobenzene: an experimental and theoretical study"
  Physica Scripta **99**, 125401 · 10.1088/1402-4896/ad8974 2024
- **E. G. F. de Miranda**, L. M. Cornetta and M. T. do N. Varella, "Low-energy electron interactions with resveratrol and resorcinol: anion states and likely dissociation pathways"
  The Journal of Physical Chemistry A **126**, 7237–7247 · 10.1021/acs.jpca.2c05789 2022

Theses

- PhD thesis: *Models for Transient Anion Dynamics*, University of São Paulo. USP digital library 2026
- Master's dissertation: *Anionic states of resveratrol and its subunits*, University of São Paulo. USP digital library 2021

Both are written in English.

Research Experience

Quantum scattering theory (2017 – present). Schwinger Multichannel Method calculations for transient anion characterization in radiosensitizing molecules (chlorobenzenes, pyrimidines, phenolic compounds). Elastic cross sections, identification of shape, Feshbach and core-excited resonances. Collaboration with experimental groups on electron transmission and mass spectrometry.

Nonadiabatic dynamics (2023 – present). Surface hopping simulations with fully correlated methods (CASPT2, MRCI, RASPT2, MRSF-TDDFT). Potential energy surfaces, nonadiabatic couplings, conical intersections. Newton-X and OpenMolcas interface development for photochemical processes and dissociative electron attachment.

Multireference quantum chemistry. Active space selection (CASSCF, RASSCF), dynamic correlation (CASPT2, RASPT2, NEVPT2), Feshbach resonances, core-excited states, strongly correlated systems.

Talks and Posters

- **Oral:** Surface Hopping for dissociation in electron-molecule collisions with correlated methods. XXXIV International Conference on Photonic, Electronic and Atomic Collisions (ICPEAC), Sapporo, Japan 2025
- **Oral:** Exploring dissociative electron attachment processes induced by vibrational Feshbach resonance through active learning and Surface Hopping. XVI Workshop on Molecular Physics and Spectroscopy (WFME), Curitiba, Brazil 2024
- **Poster:** Exploring dissociative electron attachment processes induced by vibrational Feshbach resonance through active learning and Surface Hopping. Congress of Theoretical Chemists of Latin Expression (CHITEL), Namur, Belgium 2024
- **Poster:** Low-energy electron interactions with 2,4-dichloro-5-nitropyrimidine: anion states and likely dissociation pathways. XV Workshop on Molecular Physics and Spectroscopy (WFME), São Paulo, Brazil 2023
- Anionic states of trans-resveratrol. Brazilian Symposium on Theoretical Chemistry (SBQT) 2019
- **Poster:** Estados eletrônicos da 5-cloro-uracila. Workshop on Molecular Physics and Spectroscopy (WFME) 2018
- **Poster:** Description of 5-chlorine-uracil in water. 25th USP International Symposium on Undergraduate Research (SIICUSP), selected for the international stage 2017

Teaching

Teaching assistantships at the Institute of Physics, University of São Paulo:

- Electromagnetism II (4302304), undergraduate course, Prof. Lucas Cornetta 2024
- Physics I (4302111), undergraduate course, Prof. Raul Abramo 2023

- Physics IV (4302212), undergraduate course, Prof. Márcio Varella 2022
- Introduction to Atomic and Molecular Physics (4300315), undergraduate course, Prof. Sylvio Canuto 2022
- Introduction to Solid State Physics (4300402), undergraduate course, Prof. Fernando Garcia 2021

I also give private lessons in physics, mathematics and chemistry, from middle school to graduate level, including one-on-one courses in quantum mechanics, statistical mechanics, dynamical systems and relativity, with lecture notes prepared for each student.

Grants and Awards

- FAPESP Postdoctoral Fellowship, University of Campinas (Unicamp), Grant #2026/07066-7 2026 – present
- FAPESP Research Internship Abroad (BEPE), Aix-Marseille Université, France, Grant #2023/14792-8 2024
- FAPESP PhD Fellowship, Grant #2021/09837-7 2021 – 2025
- CNPq Master's Fellowship 2019 – 2020

Event Organization

- XV Workshop on Molecular Physics and Spectroscopy (WFME), member of the organizing committee, São Paulo 2023

Technical Skills

Quantum chemistry: OpenMolcas, Gaussian, Newton-X, GAMESS, Schwinger Multichannel Method code
Methods: CASSCF/CASPT2, RASSCF/RASPT2, MRCI, (MRSF-)TDDFT, Surface Hopping, quantum scattering theory
Programming: Python (NumPy, SciPy, Matplotlib), Fortran 90/95 (MPI/OpenMP), Bash, Git, LaTeX
Computing: Linux HPC clusters, MPI/OpenMP parallelization, workflow automation

Languages

Portuguese: native • **English:** fluent • **French:** basic